

Introduction to Probabilities and Statistics

The Need for Data Analysis

Data Science

- The art of creating meaning from data

Question to ponder:

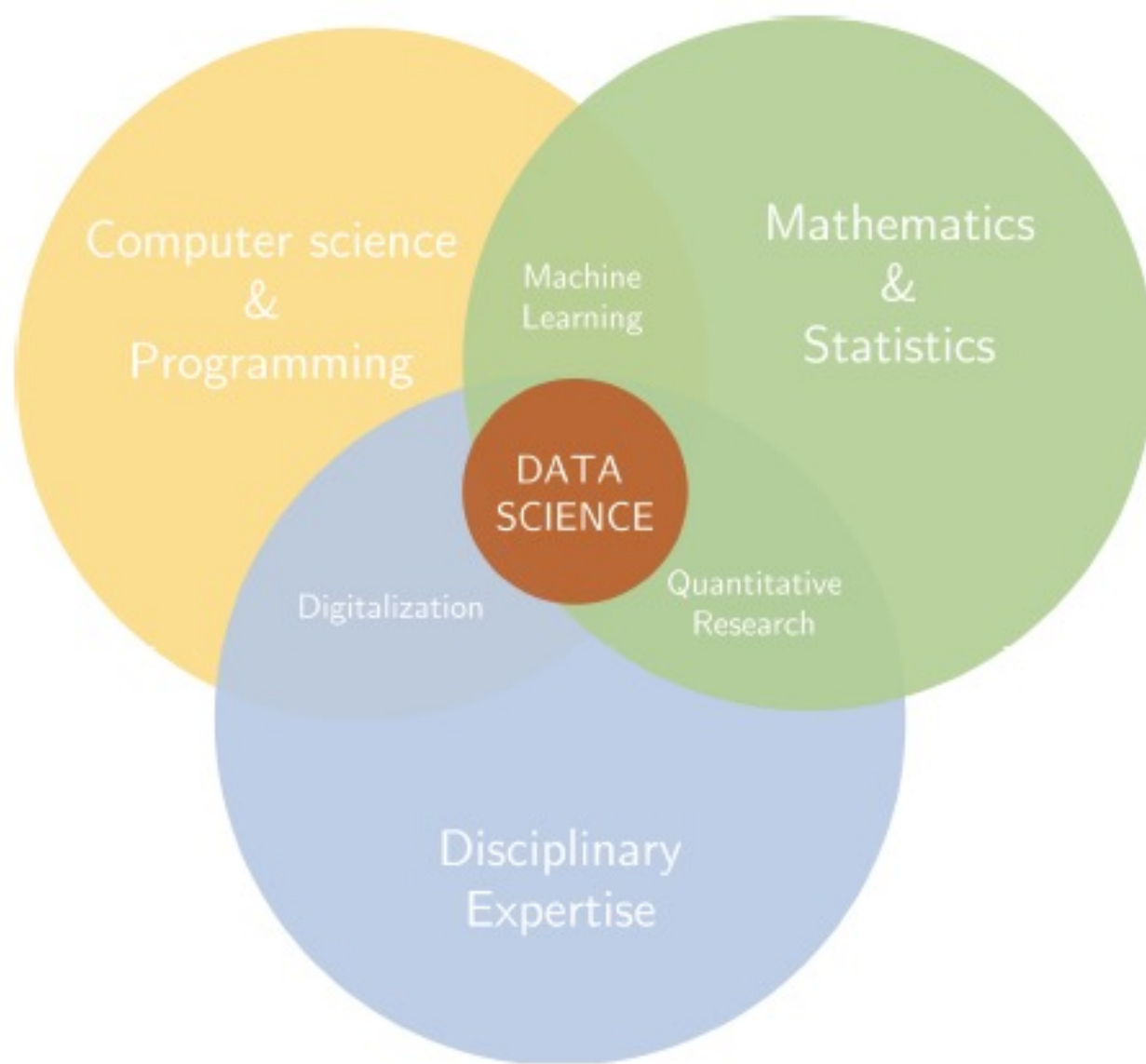
- Why do we need Probabilities and Statistics subject when we are in the Computer Science profession?

What is Data Analysis?

- Data analysis is the process of analyzing data in various formats.
- Even though data is abundant nowadays, it's available in different forms and scattered over various sources.
- Data analysis helps to clean and transform all this data into a consistent form so it can be effectively studied.
- *For data to be analyzed, we need statistical data.*

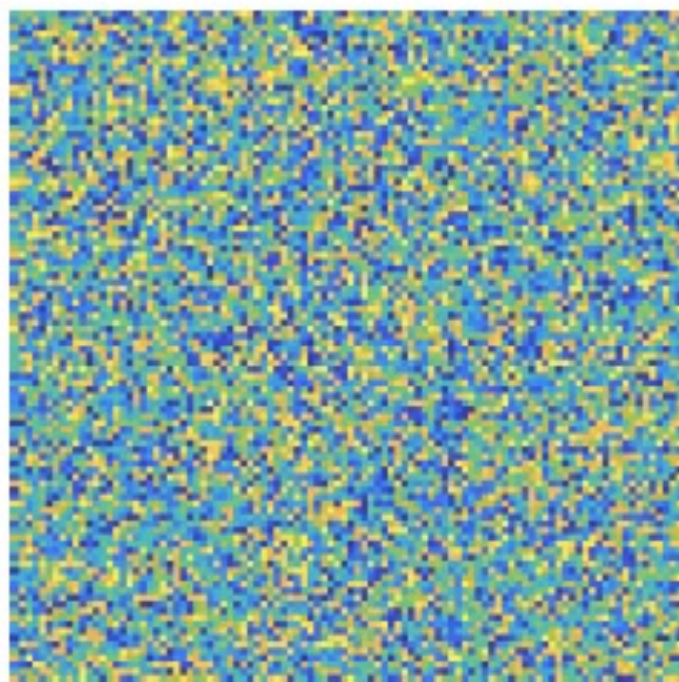
What is Data Analysis?

- Once the data is cleaned, transformed, and ready to use, it can do wonders.
- Not only does it contain a variety of useful information, studying the data collectively results in uncovering very minor patterns and details that would otherwise have been ignored.
- With the help of

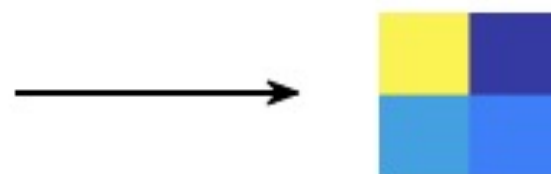


Data analysis is data reduction

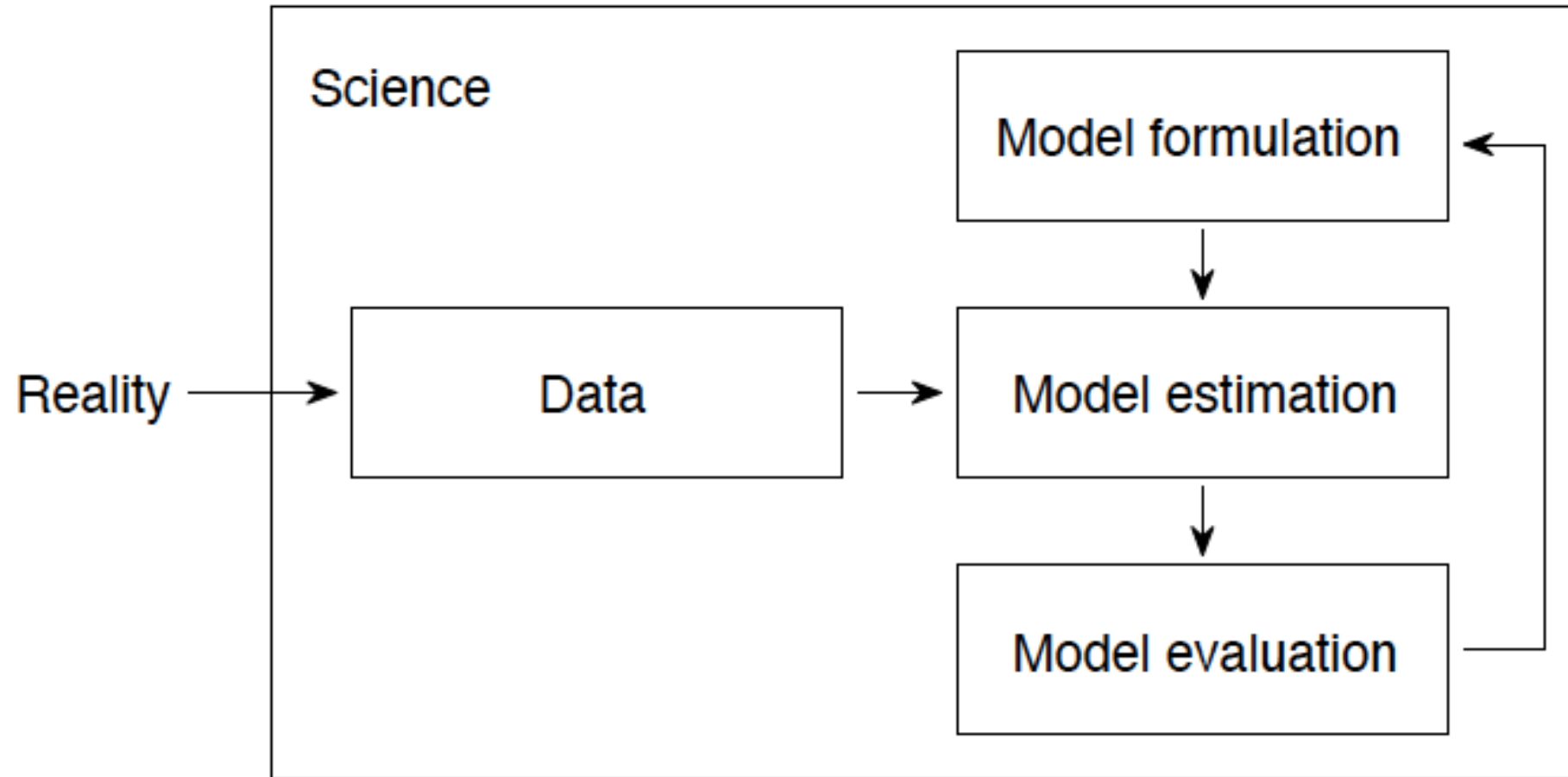
Raw Data



Reduced Data



Data analysis is model-based



Data science = Statistics = Machine learning = Artificial intelligence

Statistics

- Probabilistic models
- Theoretical analysis
- Optimality
- Asymptotics
- Science philosophy

Machine learning

- Deterministic models
- Classification
- Bayesian models
- Benchmarking
- Applications

Artificial intelligence

- Deep learning
- Reinforcement
- Decisions
- Data analysis
- Hype

What is Python?

- Python is an interpreted, object-oriented, high-level programming language with dynamic semantics.
- Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance.
- Python is a general-purpose language, which means it's designed to be used in a range of applications, including ***data science, software and web development, automation.***

What is R?

- R is a language and environment for statistical computing and graphics. It is a [GNU project](#) which is similar to the S language and environment which was developed at Bell Laboratories (formerly AT&T, now Lucent Technologies) by John Chambers and colleagues.
- R can be considered as a different implementation of S. There are some important differences, but much code written for S runs unaltered under R.
- R provides a wide variety of statistical (linear and nonlinear modelling, classical statistical tests, time-series analysis, classification, clustering, ...) and graphical techniques, and is highly extensible.